

Q12)

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

rank = 2

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{array} \right] \rightarrow \text{augmented matrix}$$

Rank = 2

Rank
Capelli
Theorem

no. of free variables = no. of variables - rank

$$= 3 - 2$$

$$= 1$$

$$x_1 + x_2 + x_3 = 0$$

$$x_1 + 2x_3 = 0$$

$$x_1 = -2x_3$$

$\therefore$ 1 free variable.

$$x_2 - 2x_3 + x_3 = 0$$

$$x_2 = +x_3$$

$$x_1 = -2x_3$$

$$x_2 = x_3$$

$$x_3 = x_3$$

put $x_3 = k$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = k \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore$ line